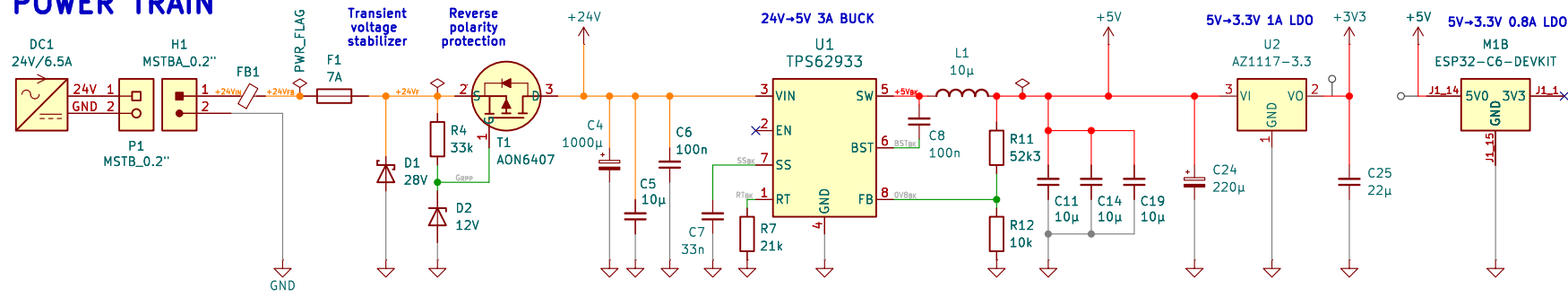
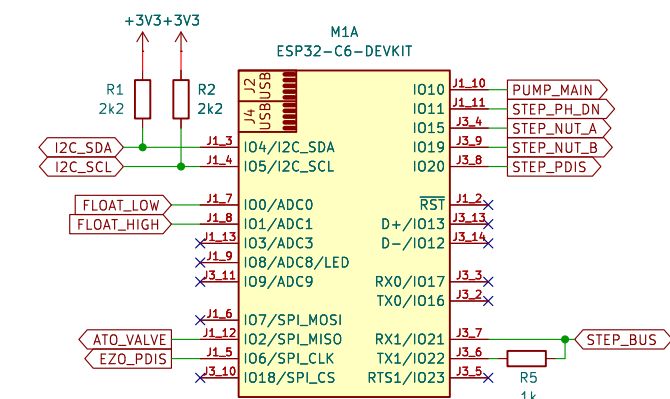


POWER TRAIN

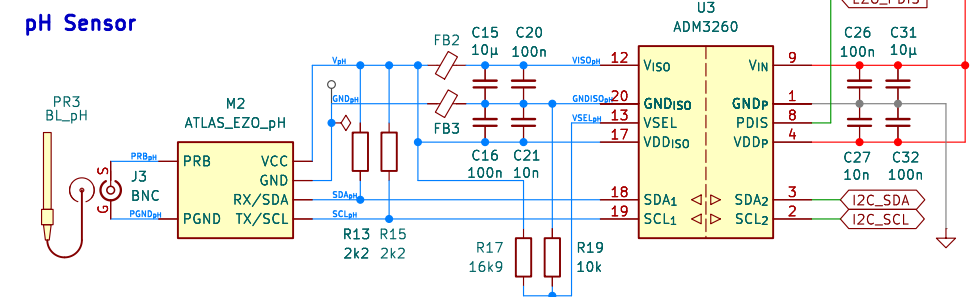


SoC

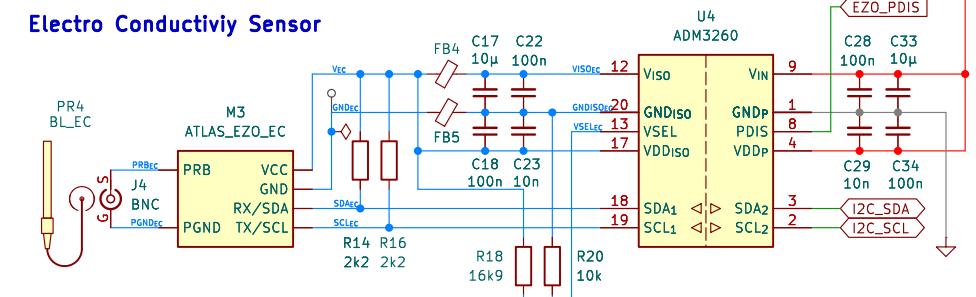


I2C SENSORS

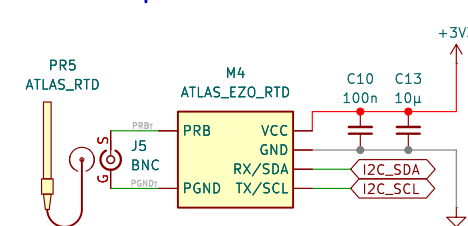
pH Sensor



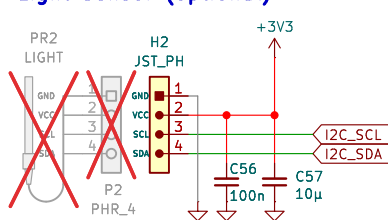
Electro Conductivity Sensor



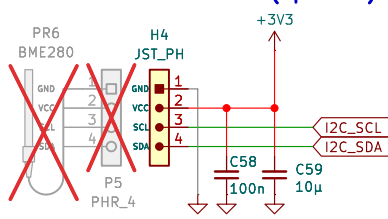
Water Temperature Sensor



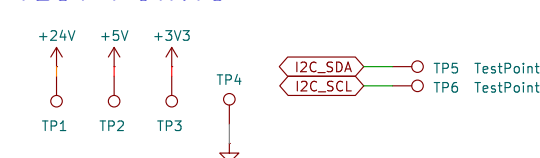
Light Sensor (optional)



Environmental Sensor (optional)

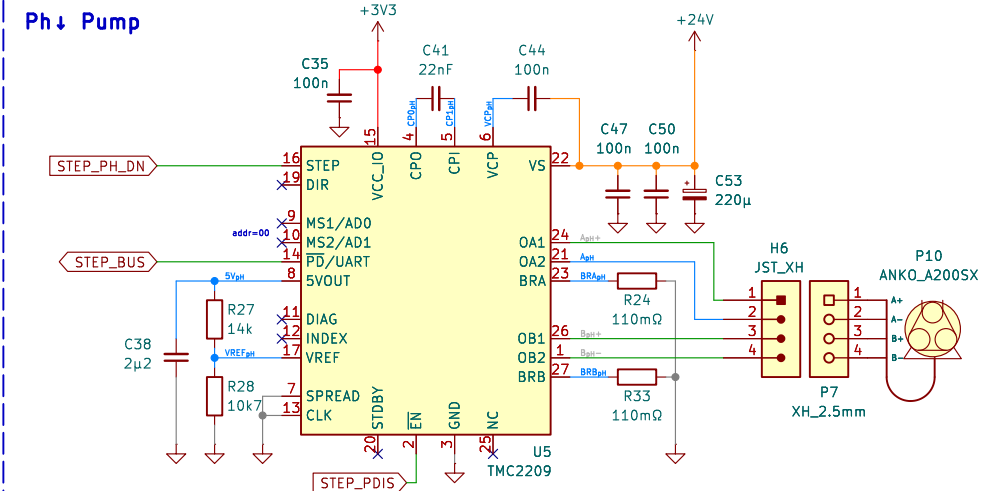


TEST POINTS

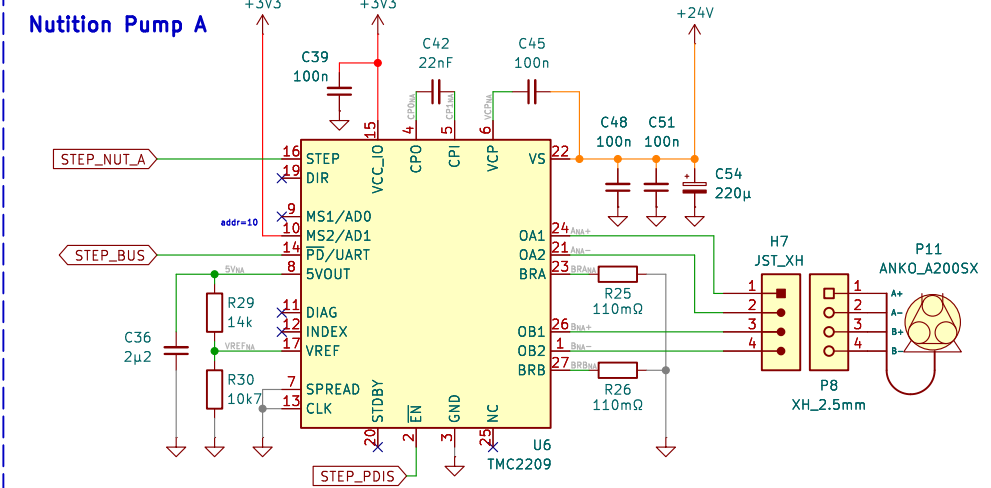


PERISTALTIC PUMP DRIVERS

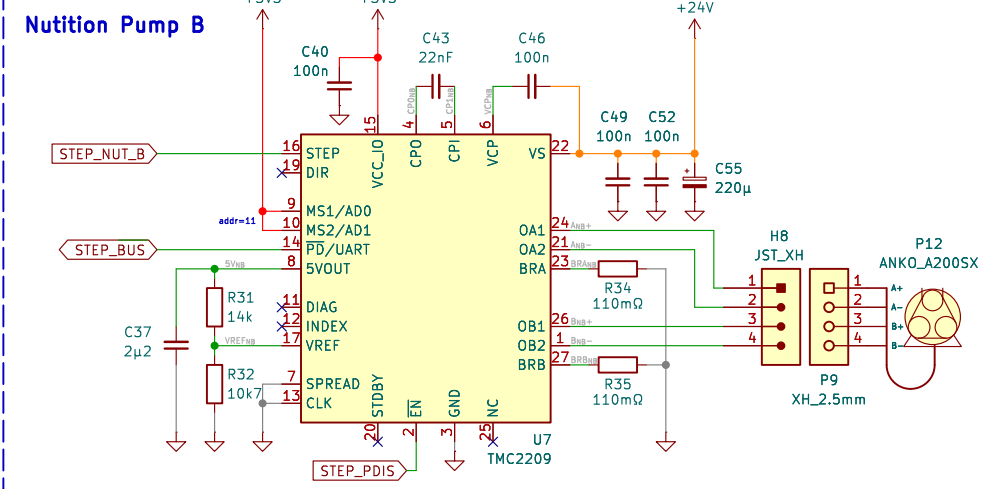
Ph Pump



Nutrition Pump A

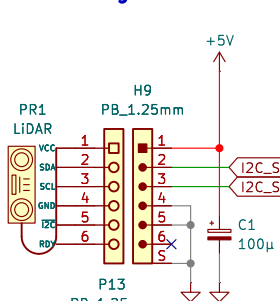


Nutrition Pump B

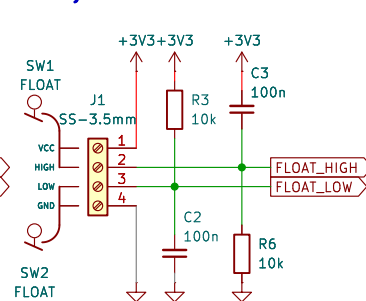


WATER LEVEL SENSOR/SWITCHES

Water Height Sensor

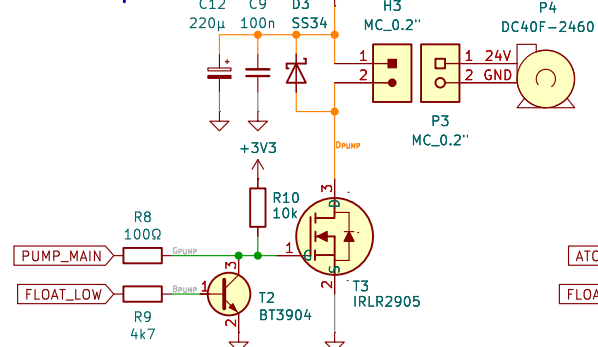


Safety Switches

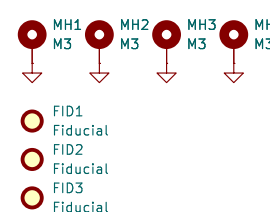
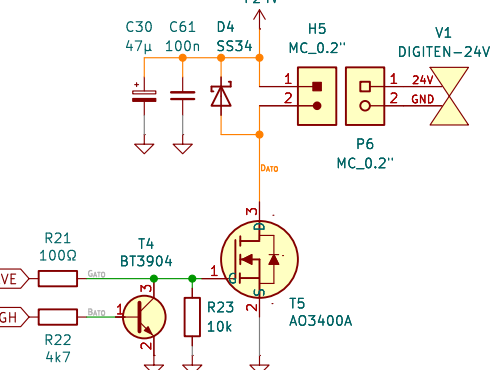


MAIN PUMP & VALVE DRIVERS

Main Pump



ATO Valve



Autonomous, data-rich hydroponics irrigation automation with precision control and remote monitoring. ESP32-C6 based with full sensor suite, dosing pump control, and automatic top-off (ATO) capability. Designed for Home Assistant or Matter over Thread/WiFi.
<https://github.com/cvnon/OPNhydro>
 Sheet:
 File: OPNhydro_r2.kicad_sch
Title: OPNhydro
 Size: A3 Date: 2026-03-03 Rev: r2
 KiCad E.D.A. 9.0.7 Id: 1/1